1 Real and Complex Numbers

Recall that the operations of addition and multiplication of real numbers satisfy the following properties:

Properties of Addition
(A1) (Commutativity) $x+y=y+x$ for all $x,y\in\mathbb{R}$.
(A2) (Associativity) $(x+y)+z=x+(y+z)$ for all $x,y,z\in\mathbb{R}$
(A3) (Identity) There exists an element $0\in\mathbb{R}$ such that $0+x=x$ for every $x\in\mathbb{R}$.
(A4) (Additive Inverses) To each $x\in\mathbb{R}$ there exists an element $-x\in\mathbb{R}$ such that $x+(-x)=0$.

Properties of Multiplication
(M1) (Commutativity) $xy=yx$ for all $x,y\in\mathbb{R}$.
(M2) (Associativity) $(xy)z=x(yz)$ for all $x,y,z\in\mathbb{R}$.
(M3) (Identity) There exists a non-zero element $1\in\mathbb{R}$ such that $1x=x$ for every $x\in\mathbb{R}$.
(M4) (Multiplicative inverses of nonzero elements) For each non-zero element $x\in\mathbb{R}$, there exists an element $x^{-1}\in\mathbb{R}$ (also denoted as $1/x$) such that $xx^{-1}=1$.

Distributive Property
(D) $x(y+z)=xy+xz$ for all $x,y,z\in\mathbb{R}$.

These properties are also satisfied by the addition and multiplication of *complex numbers*. Recall that a complex number, z, is an ordered pair of real numbers $z=(a,b)$, which we write as

$$z=a+bi,$$

The real numbers a and b are called the *real* and *imaginary parts* of z, respectively. Two complex numbers $a+bi$ and $c+di$ are *equal* if and only if $a=c$ and $b=d$.

Addition and multiplication of complex numbers $z=a+bi$ and $w=c+di$ are defined by

$$\begin{aligned} z+w&=(a+bi)+(c+di)=(a+c)+(b+d)i,\\ zw&=(a+bi)(c+di)=(ac-bd)+(bc+ad)i. \end{aligned}$$

Note that there is a 1-1 correspondence between elements of $\mathbb{R}$ with elements of $\mathbb{C}$ the form $a+0i$. Moreover, for such elements, the operations above reduce to the usual operations on real numbers, in the sense that

$$\begin{aligned} a+0i+b+0i&=(a+b)+0i\\ (a+0i)(b+0i)&=ab+0i. \end{aligned}$$

In this way we identify $\mathbb{R}$ as a subset of $\mathbb{C}$ and write a for $a+0i$. We will also write $0+bi$ as just bi and $0+1i$ as just i. Note that the definition of multiplication above implies that

$$i^2=-1.$$

The fact that addition and multiplication of complex numbers satisfies the properties above follows from the fact that they are satisfied by addition and multiplication of real numbers.

Example 1.1. To prove that $zw = wz$ for all $z, w \in \mathbb{C}$, write $z = x + iy$ and $w = u + iv$. Then

$$\begin{aligned} zw &= (x+iy)(u+iv) \\ &= (xu - yv) + i(yu + xv) && \text{(definition of multiplication in } \mathbb{C}) \\ &= (ux - vy) + i(uy + vx) && \text{(commutativity of multiplication in } \mathbb{R}) \\ &= wz && \text{(definition of multiplication in } \mathbb{C}) \end{aligned}$$

This proves that (M1) holds for multiplication in $\mathbb{C}$.

In particular, if $b \in \mathbb{R}$, then $bi = ib$, so we can write a complex number as $a + bi$ or $a + ib$.

The other properties are proved similarly: one writes out each complex number in terms of its real and imaginary parts, and then uses the fact that the corresponding property holds for real numbers.

Remark 1.2 (Digression on Fields). More generally, any set F, which has two operations, each taking a pair of elements of F to another element of F, which satisfy properties (A1)-(A4), (M1)-(M4),(D), is called a *field*. These properties are called the *field axioms*. Thus, $\mathbb{R}$ and $\mathbb{C}$ are fields, as is $\mathbb{Q}$, the set of rational numbers (together with its operations of addition and multiplication). The integers, $\mathbb{Z}$, is not a field - the element 2, for example, does not have a multiplicative inverse in $\mathbb{Z}$. Another example of a field is the set $\{0, 1\}$ with the usual operations of addition and multiplication, except that $1 + 1$ is defined to be equal to 0.

Many definitions, theorems, and proofs in linear algebra that work for the fields $\mathbb{R}$ and $\mathbb{C}$ also work for arbitrary fields. While we will not be dealing with fields other than $\mathbb{R}$ or $\mathbb{C}$ in this course, since some IMSA students in recent years have completed research projects involving linear algebra over finite fields such as the one mentioned above, [1] this seemed worth pointing out.

So that we can conveniently make definitions and prove theorems that apply to both real and complex numbers,

Going forward we will let F stand for either $\mathbb{R}$ or $\mathbb{C}$.

[1] See https://digitalcommons.imsa.edu/sir_presentations/2022/session3/9/ and https://digitalcommons.imsa.edu/sir_presentations/2024/session2/18/.

2 Implications of the Field Axioms

Many other properties follow from the field axioms. In any field F, the additive identity of (A3) is necessarily unique. To prove this, suppose both $0, 0' \in F$ satisfy $0 + x = x$ for all $x \in F$. Then

$$\begin{aligned} 0 &= 0' + 0 && \text{(by (A3), since } 0' \text{ is an additive identity)} \\ &= 0 + 0' && \text{(by (A1))} \\ &= 0' && \text{(by (A3), since 0 is an additive identity).} \end{aligned}$$

1. Prove the following, justifying each step as above:

   (a) The multiplicative identity, 1, of (M3), is unique.
   Suppose $1, 1'$ are both satisfy $1x = x$ for all $x \in F$. Then

   $$\begin{aligned} 1 &= 1'1 && \text{(by (M3), since } 1' \text{ is a multiplicative inverse)} \\ &= 11' && \text{(by (M1))} \\ &= 1' && \text{(by (M3), since 1 is a multiplicative inverse)} \end{aligned}$$

   (b) The additive inverse of (A4) of $x \in F$ is unique. This justifies the notation "$-x$".
   Suppose $x + y = 0$ and $x + z = 0$. Then

   $$\begin{aligned} y &= 0 + y && \text{by (A3)} \\ &= (x + z) + y && \text{by (A4), since } z \text{ is an additive inverse for } x \\ &= (z + x) + y && \text{by (A1)} \\ &= z + (x + y) && \text{by (A2)} \\ &= z + 0 && \text{by (A4), since } y \text{ is an additive inverse for } x \\ &= 0 + z && \text{by (A1)} \\ &= z && \text{by (A3)} \end{aligned}$$

   (c) The multiplicative inverse of (M4) of $x \in F \setminus \{0\}$ is unique. This justifies the notation "x^{-1}".
   Suppose $yx = 1$ and $zx = 1$. Then

   $$\begin{aligned} y &= 1y && \text{(by (M3))} \\ &= (zx)y && \text{(by (M4), since } z \text{ is a multiplicative inverse for } x) \\ &= z(xy) && \text{(by (M2))} \\ &= z(yx) && \text{(by (M1))} \\ &= z1 && \text{(by (M4), since } y \text{ is a multiplicative inverse for } x) \\ &= 1z && \text{(by (M1))} \\ &= z && \text{(by (M3))} \end{aligned}$$

2. Prove the following, justifying each step.

   (a) If $x+y=x+z$, then $y=z$. This is called the *(additive) cancellation law*.
   Adding $-x$ to the left of both sides of the equation gives

$$\begin{aligned} -x+(x+y) &= -x+(x+z) & (-x+x)+y &= (-x+x)+z \quad \text{(by (A2))} \\ (x+(-x))+y &= (x+(-x))+z \quad \text{(by (A1))} \\ 0+y &= 0+z \quad \text{(by (A4))} \\ y &= z \quad \text{(by (A3))} \end{aligned}$$

   (b) $-(-x)=x$
   By definition, $-(-x)$ is an additive inverse of $-x$. However, x is also an additive inverse of $-x$. By uniqueness of additive inverses, $-(-x)=x$.

   (c) If $x\neq 0$ and $xy=xz$ then $y=z$. This is also called the *(multiplicative)* cancellation law.
   The proof is almost identical to the one above.

   (d) If $x\neq 0$ then $1/(1/x)=x$.
   The proof is almost identical to the one above.

3. Prove the following, justifying each step.

   (a) $0x=0$ (*Hint*: $0x=(0+0)x$)

$$\begin{aligned} 0x &= (0+0)x \quad \text{(by (A3))} \\ 0x &= 0x+0x \quad \text{(by (D))} \\ 0x+0 &= 0x+0x \quad \text{(by (A3))} \\ 0 &= 0x \quad \text{(by the cancellation law)} \end{aligned}$$

   (b) If $x\neq 0$ and $y\neq 0$ then $xy\neq 0$. (*Hint*: Suppose $xy=0$ and apply part (a) to reach a contradiction.)
   Suppose $xy=0$. Then

$$1=\left(\frac{1}{y}\right)\left(\frac{1}{x}\right)xy=\left(\frac{1}{y}\right)\left(\frac{1}{x}\right)0=0,$$

   a contradiction. Hence, $xy\neq 0$.

   (c) $(-x)y=-(xy)=x(-y)$ (*Hint*: Proving the first equality is equivalent to proving that $(-x)y+xy=0$. Similarly for the second.)

$$\begin{aligned} (-x)y+xy &= y(-x)+yx \\ &= y(-x+x) \\ &= y0 \\ &= 0 \quad \text{(by part (a)).} \end{aligned}$$

Similarly,

$$\begin{aligned} x(-y)+xy &= x(-y+y) \\ &= x0 \\ &= 0 \qquad \text{(by part (a)).} \end{aligned}$$

(d) $(-x)(-y) = xy$ (*Hint*: Use part (c).)

$$\begin{aligned} (-x)(-y) &= -(x(-y)) \qquad \text{(by the first equality in part (c), since } -(-x) = x\text{)} \\ &= -(-(xy)) \qquad \text{(by the second equality in part (c), since } -(-y) = y\text{)} \\ &= xy. \end{aligned}$$

4. (a) In $\mathbb{C}$, what is 0? 1? Justify your answers.
$0 = 0+0i$, $1 = 1+0i$ are easily checked to satisfy (A3) and (M3), respectively. By problem 1 above, these are unique.

   (b) Given $z = a+bi \in \mathbb{C}$, what is $-z$? z^{-1}? Justify your answers.
$-z = -a-bi$, $\quad z^{-1} = \frac{a-bi}{a^2+b^2}$ are easily checked to satisfy (A4) and (M4), respectively. By problem 1 above, these are unique.